HDOC Day-45 Assignment

Question

Write a menu-driven C program to perform the following operations using separate functions:

- Change the date format from dd/mm/yyyy to dd-mmm-yy. E.g. Input: 12/10/2026, Output: 12-Oct-26
- Print all sub-strings of a string. E.g. Input: abc, Output: a, ab, abc, b, bc, c

1. Problem Understanding

The program is menu-driven and performs two operations using separate functions. It can convert a date from dd/mm/yyyy format to dd-mmm-yy format and print all possible contiguous sub-strings of a given string. The program uses simple arrays, loops, functions, and a switch statement so that the logic remains clear and beginner-friendly.

2. Standard Test Cases

S.No.	All Inputs	Expected Output
1	Choice = 1 Date: 12/10/2026	12-Oct-26
2	Choice = 1 Date: 01/01/2025	01-Jan-25
3	Choice = 2 String: abc	a, ab, abc, b, bc, c
4	Choice = 2 String: ab	a, ab, b
5	Choice = 3	Program exits successfully.

3. Algorithm

1. Start.
2. Display the menu containing the two operations and Exit.
3. Read the user choice.
4. For choice 1, read the date in dd/mm/yyyy format and call changeDateFormat().
5. Separate the day, month, and year values from the input date.
6. Use the month number to select the corresponding three-letter month name.
7. Display the converted date in dd-mmm-yy format.
8. For choice 2, read a string and call printSubstrings().
9. Use nested loops to select every starting position and every possible ending position.
10. Print the characters between the selected positions to display each contiguous sub-string.
11. For choice 3, exit the program. For any other choice, display an invalid-choice message.
12. Stop.

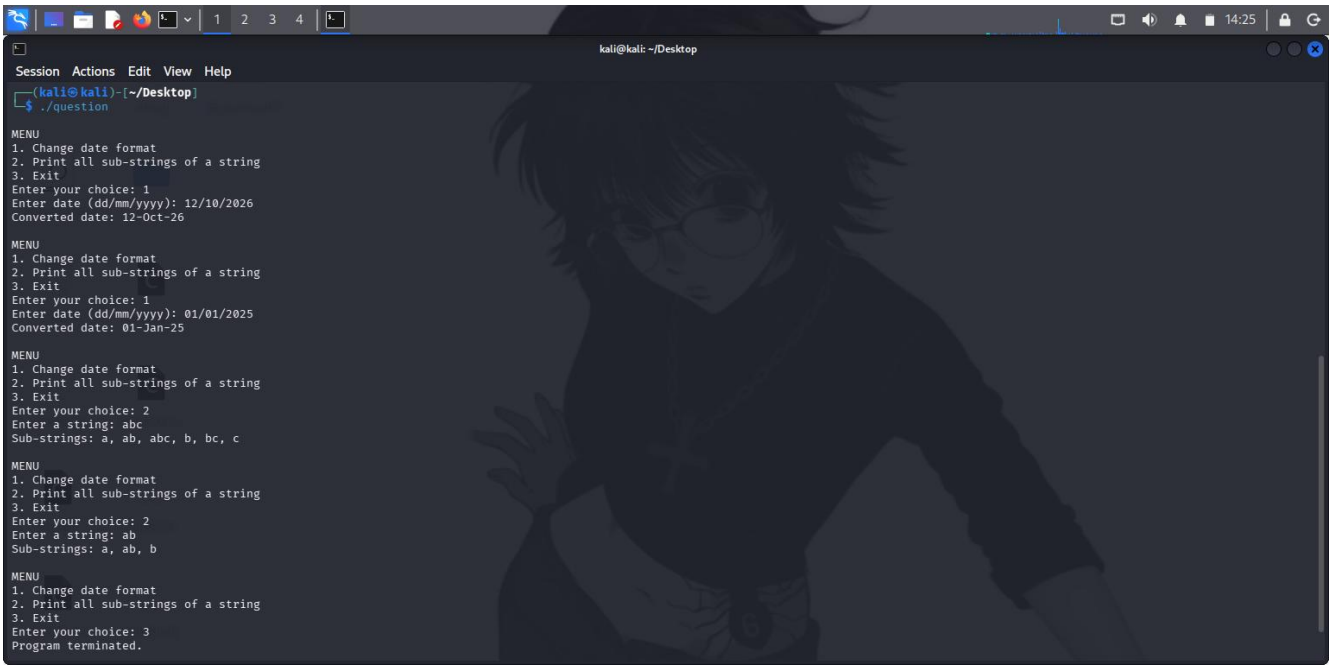
4. C Program

```
#include <stdio.h>
#include <string.h>
void changeDateFormat()
{
   int day, month, year;
```

```
  char monthName[12][4] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov",
"Dec"};
  printf("Enter date (dd/mm/yyyy): ");
  scanf("%d/%d/%d", &day, &month, &year);
  if (month >= 1 && month <= 12)
    printf("Converted date: %02d-%s-%02d\n", day, monthName[month - 1], year % 100);
  else
    printf("Invalid month.\n");
}
void printSubstrings()
{
  char str[100];
  int i, j, k, length;
  printf("Enter a string: ");
  scanf(" %99[^\n]", str);
  length = strlen(str);
  printf("Sub-strings: ");
  for (i = 0; i < length; i++)
  {
    for (j = i; j < length; j++)
    {
      for (k = i; k <= j; k++)
        printf("%c", str[k]);
      if (!(i == length - 1 && j == length - 1))
        printf(", ");
    }
  }
  printf("\n");
}
int main()
{
  int choice;
  do
  {
    printf("\nMENU\n");
    printf("1. Change date format\n");
    printf("2. Print all sub-strings of a string\n");
```

```
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
      case 1:
        changeDateFormat();
        break;
      case 2:
        printSubstrings();
        break;
      case 3:
        printf("Program terminated.\n");
        break;
      default:
        printf("Invalid choice.\n");
    }
  }
  while (choice != 3);
  return 0;
}
```

5. Compilation and Execution Evidence



6. Verification of All Test Cases

Test	Input Summary	Actual Output	Result
1	12/10/2026	Converted date: 12-Oct-26	Pass
2	01/01/2025	Converted date: 01-Jan-25	Pass
3	abc	a, ab, abc, b, bc, c	Pass
4	ab	a, ab, b	Pass
5	Choice = 3	Program terminated.	Pass